📝 Factory-to-Customer Shipping Route Efficiency Analysis for Nassau Candy Distributor

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📌 Abstract

In modern supply chain systems, shipping efficiency plays a critical role in ensuring customer satisfaction and operational performance. This project analyses the factory-to-customer shipping routes of Nassau Candy Distributor to identify inefficiencies, delays, and bottlenecks across regions. Using transactional order data, key metrics such as shipping lead time, delay frequency, and route efficiency scores were computed. The study reveals significant variations in delivery performance across regions and shipping modes. The findings provide actionable insights to optimize logistics operations, reduce delays, and improve delivery reliability.

📖 1. Introduction

Efficient logistics management is a cornerstone of successful distribution systems. Companies operating at a national scale must ensure timely delivery while maintaining cost efficiency. Nassau Candy Distributor ships products from multiple factories to customers across various regions, making route optimization essential.

However, without route-level performance visibility, logistics decisions often remain reactive. This project aims to provide a data-driven framework to evaluate shipping efficiency and identify improvement opportunities.

❗ 2. Problem Statement

The organization lacks clarity on:

- Efficient vs inefficient shipping routes
- Regions experiencing consistent delays
- Impact of shipping modes on delivery performance
- Geographic bottlenecks affecting logistics

This lack of insight limits the ability to optimize operations and reduce delivery delays.

⚙️ 3. Methodology

The analysis followed a structured data analytics pipeline:

🔹 Data Cleaning & Validation

- Converted date fields into standard datetime format
- Removed invalid or unrealistic lead times
- Handled missing shipment records
- Standardized geographic fields

🔹 Feature Engineering

- Calculated **Shipping Lead Time** (Ship Date − Order Date)
- Mapped products to factories
- Defined routes:
  - Factory → Region
  - Factory → State

🔹 Aggregation & Analysis

- Grouped shipments by routes and ship modes
- Calculated:
  - Total shipments
  - Average lead time
  - Lead time variability

🔹 KPI Development

- Delay Frequency
- Route Efficiency Score
- Average Lead Time

📊 4. Exploratory Data Analysis (EDA)

EDA revealed key patterns:

- Most shipments are completed within a short time range after data cleaning
- Significant variation exists in lead times across regions
- Certain routes consistently perform better than others
- Shipping mode influences delivery speed significantly

🔍 5. Key Insights

1. Route Performance Variation

Some factory-to-state routes consistently exhibit lower lead times, indicating efficient logistics networks.

2. Regional Bottlenecks

Certain regions show higher average lead times, suggesting possible infrastructure or distance-related challenges.

3. High Volume, High Delay Routes

Routes with both high shipment volume and high delays represent critical bottlenecks.

4. Shipping Mode Impact

Expedited shipping methods significantly reduce delivery time compared to standard shipping.

🌍 6. Geographic Bottleneck Analysis

Analysis identified:

- Regions with high average lead times
- States experiencing frequent delays
- Areas where shipment volume is high but performance is poor

These findings highlight locations where logistics optimization is most needed.

🚚 7. Ship Mode Performance Analysis

Shipping modes were categorized into:

- Standard Shipping
- Expedited Shipping

Findings:

- Expedited shipping delivers faster but may incur higher costs
- Standard shipping is widely used but has higher lead times
- Choosing the right shipping mode depends on urgency vs cost

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📈 8. Key Performance Indicators (KPIs)

🔹 Shipping Lead Time

Measures delivery duration

🔹 Average Lead Time

Average delivery time per route

🔹 Route Volume

Number of shipments per route

🔹 Delay Frequency

Percentage of shipments exceeding threshold

🔹 Route Efficiency Score

Normalized metric to rank route performance

These KPIs enable continuous monitoring of logistics performance.

💡 9. Recommendations

Based on the analysis:

✔ Optimize High-Delay Routes

Focus on routes with high delay frequency and high shipment volume

✔ Introduce Regional Distribution Centers

Reduce distance-related delays

✔ Use Expedited Shipping Strategically

Apply for high-priority or delay-prone shipments

✔ Improve Route Planning

Leverage data insights to redesign inefficient routes

🔮 10. Future Scope

- Predictive modeling for delivery time estimation
- Distance-based route optimization
- Real-time logistics monitoring dashboard

🏁 10. Conclusion

This project provides a comprehensive, data-driven analysis of shipping route efficiency for Nassau Candy Distributor. By transforming raw shipment data into actionable insights, the organization can proactively address logistics inefficiencies, reduce delivery delays, and enhance customer satisfaction.

The approach demonstrates how data analytics can significantly improve supply chain decision-making and operational performance.